



Loan Rangers

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Problem Space

kaggle



HOME CREDIT GROUP · FEATURED PREDICTION COMPETITION · 6 YEARS AGO

Home Credit Default Risk

Can you predict how capable each applicant is of repaying a loan?

Dataset

Tranche 1

application_{train|test}.csv

- Main tables – our train and test samples
- Target (binary)
- Info about loan and loan applicant at application time

Tranche 2

bureau.csv

- Application data from previous loans that client got from other institutions and that were reported to Credit Bureau
- One row per client's loan in Credit Bureau

previous_application.csv

- Application data of client's previous loans in Home Credit
- Info about the previous loan parameters and client info at time of previous application
- One row per previous application

Tranche 3

bureau_balance.csv

- Monthly balance of credits in Credit Bureau
- Behavioral data

POS_CASH_balance.csv

- Monthly balance of client's previous loans in Home Credit
- Behavioral data

instalments_payments.csv

- Past payment data for each installments of previous credits in Home Credit related to loans in our sample
- Behavioral data

credit_card_balance.csv

- Monthly balance of client's previous credit card loans in Home Credit
- Behavioral data



Cleaning, Encoding, Selecting and Resampling Data

Encoding Data: Performed 1-hot encoding to turn all data into numerical values.

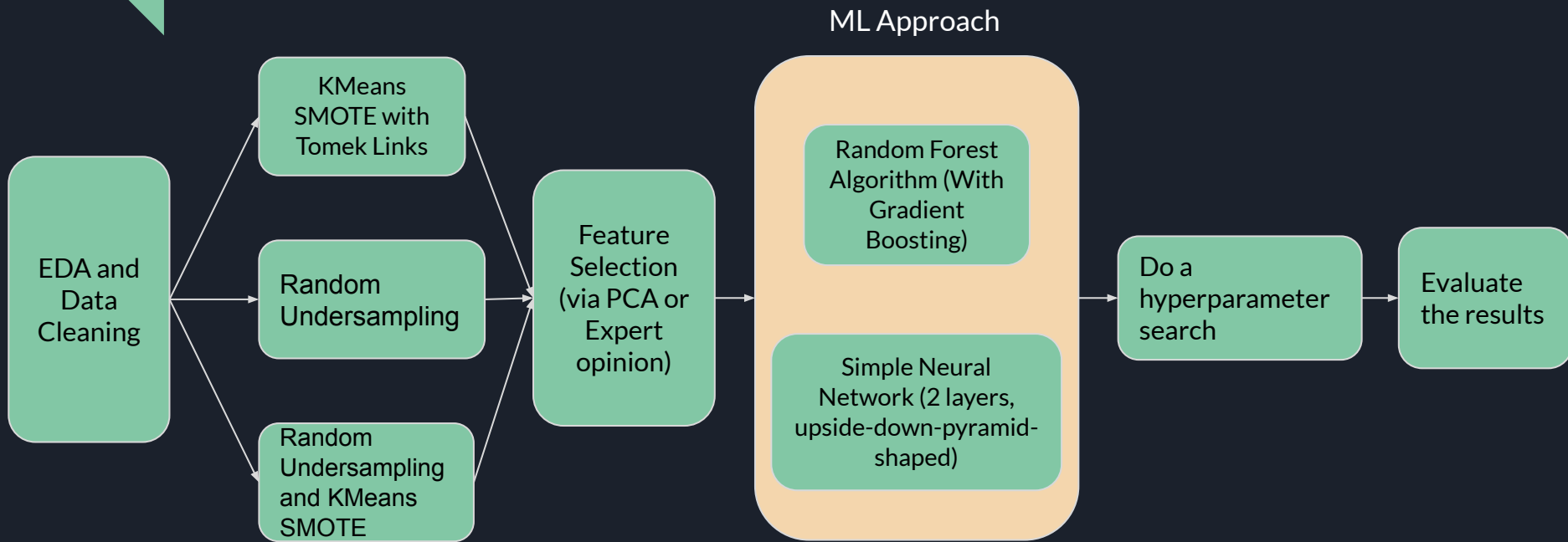
Encoding Data from Secondary Tables: Computed the mean, median and variance of the samples mapped to the main dataset to try and capture the trend from multiple samples.

Selecting Data: Performed a Principal Component Analysis to identify the components that explained the most variance

Resampling Data: Performed Random Undersampling, KMeans Synthetic Minority Oversampling Technique (SMOTE) and a combination of both to rebalance classes

Dataset/Dimensions	Tranche 1	Tranche 2	Tranche 3
Number of Features:	250	795	1266
Training	0.0: 167912, 1.0: 14806 Tot: 182718	0.0: 167912, 1.0: 14806 Tot: 182718	0.0: 167912, 1.0: 14806 Tot: 182718
KMeans SMOTE with Tomek Links	0.0: 167910, 1.0: 91891 Tot: 259801	0.0: 167909, 1.0: 94953 Tot: 259801	0.0: 167905, 1.0: 82803 Tot: 259801
Random Undersampling	0.0: 14806, 1.0: 14806	0.0: 14806, 1.0: 14806	0.0: 14806, 1.0: 14806
	Tot: 29612	Tot: 29612	Tot: 29612
Random Undersampling and KMeans SMOTE	0.0: 74030, 1.0: 14806 Tot: 88836	0.0: 74030, 1.0: 14958 Tot: 88988	0.0: 74030, 1.0: 15001 Tot: 89031
Test	0.0: 55976, 1.0: 4930 Tot: 60906	0.0: 55976, 1.0: 4930 Tot: 60906	0.0: 55976, 1.0: 4930 Tot: 60906
Validation	0.0: 55976, 1.0: 4931 Tot: 60907	0.0: 55976, 1.0: 4931 Tot: 60907	0.0: 55976, 1.0: 4931 Tot: 60907

Approach



Results

1. Random Forest Algorithm (With Gradient Boosting)

	Tranche 1	Tranche 2	Tranche 3
Training	0.65550	0.64966	0.61137
KMeans SMOTE with Tomek Links	0.62566 0.62880	0.62694 0.63794	0.63791
Random Undersampling	0.65537	0.64966	0.61137
Random Undersampling and KMeans SMOTE	0.71708 0.72400	0.67342	0.61361

HyperParameter-Optimized Score is highlighted

2. Simple Neural Network (2 layers, upside-down-pyramid-shaped)

	Tranche 1	Tranche 2	Tranche 3
Training	0.69650	0.69843	0.70503
KMeans SMOTE with Tomek Links	0.65663	0.66357	0.64474
Random Undersampling	0.69232	0.68897	0.70134
Random Undersampling and KMeans SMOTE	0.69404	0.70722	0.70098



Future Extensions

Feature Engineering:

- Embed more useful information using automated feature tools

Efficient Hyperparameter Tuning:

- Leverage Bayesian optimization or evolutionary algorithms

Monetary Risk Integration:

- Incorporate financial metrics (expected returns, losses) for holistic decision-making
- Align predictive risk scores with economic outcomes for more actionable insights